4. Objects and Classes

Object:

An object in java is the physical as well as a logical entity. An entity that has state and behaviour is known as an object.

Ex: chair, bike, pen, etc.

An object is the instance of a class. An object has three characteristics.

- state: Represents data of an object
- Behaviour: Represents behaviour (functionality) of an object.
- Identify: It is used internally by JVM to identify each object uniquely.

class:

A class is a group of objects which have common properties. It is a template or blueprint from which objects are created. It is the logical entity. Class is a user defined blueprint or prototype from which objects are created.

A class in Java contains:

- fields
- methods
- constructors
- Blocks
- Nested class and interface

Syntax to declare a class:

```
class <class_name>{
    field;
    method;
}
```

Date :

Instance variable in Java:

A variable which is created inside the class but outside the method is known as an instance variable. Instance variable doesn't get memory at compile time. It gets memory at runtime when an object or instance is created. That is why it is known as an instance variable.

Method in java:

In Java, a method is like a function which is used to expose the behaviour of an object

Advantage of Method:

- code reusability
- code Optimization

New keyword in Java:

The new keyword is used to allocate memory at runtime. All objects get memory in Heap memory area.

Object and Class Example: main within the class:

Here, we are creating a main() method inside the class

```
class Student{
int id;
string name;
public static void main (string args[]){
 Student st = new Student ();
```

Date :

```
System.out.println(s1.id);
System.out.println(s1.name);
}
}
```

Output :

```
0
null
```

Object and Class Example : main outside the class

```
class student{
int id;
string name;
}
class TestStudent1{
public static void main(String args[]){
Student s1 = new student();
System.out.println(s1.id);
System.out.println(s1.name);
}
}
```

3 ways to initialize object :

There are 3 ways to initialize object in Java.

1. By reference variable
2. By method
3. By constructor

Date :

1) Object and class Example: Initialization through reference

Initializing an object means storing data into the objec
Let's see a simple example.

```
class student{
int id;
String name;
}
class TestStudent2{
  public static void main (String args[]){
    Student s1 = new Student();
    s1.id = 101;
    s1.name = "Sonoo";
    System.out.println (s1.id+ " "+s1.name); //printing member
  }
}
```

output:

```
101 Sonoo
```

2) Object and class Example: Initialization through method

In this example, we are creating the two objects of Student class and initializing the value to these objects by invoking the insertRecord method. Here, we are displaying the state(data) of the objects by invoking the displayInformation() method.

```
class student{
int rollno;
```

Date

```
string name;
void insertRecord (int r, string n){
  roll_no = r;
  name = n;
}
void displayInformation (){
  system.out.println (rollno + " " + name);}
}
class TestStudent4{
  public static void main (string args[]){
  student s1 = new student ();
  student s2 = new student ();
  s1.insertRecord (111, "karan");
  s2.insertRecord (222, "Aryan");
  s1. displayInformation ();
  s2. displayInformation ();
  }
}
```

Output:

```
111 karan
222 Aryan
```